

An Analysis of Yield Tokenization and Arbitrage Opportunities for Liquid Restaking Tokens on Pendle

Faris Chaudhry **Jian-Ren Chen**
faris.chaudhry@nethermind.io jian.chen@nethermind.io

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Abstract

Pendle is a protocol to split yield-bearing tokens into principal and yield components, providing users to leverage and trade yield. This paper analyses liquid restaking tokens (LRTs) on Pendle, addressing five primary considerations: the drivers of fixed yields, correlations between different LRTs, correlations across various tenors, the impact of liquidity on yields, and potential arbitrage strategies. We analyse 12 different contracts, considering a range of liquidity, tokens, and maturities. Using principal component analysis (PCA), correlation analysis, and random forest regression, we explore the significant drivers of fixed yields and interactions between YT and PT prices. Our findings reveal a high correlation between fixed yields and implied APYs, an illiquidity factor leading to some LRTs having higher APYs than others, and that, while principal tokens align closely with theoretical prices, yield tokens exhibit market inefficiencies due to varying rewards and protocol-specific idiosyncrasies. Additionally, we backtest several arbitrage strategies based on these findings.

1 Introduction

1.1 Overview of Yield Tokenization

In traditional finance, interest rate derivatives represent one of the largest markets, with over \$700 trillion of outstanding notional value in 2023, due to their extensive use in hedging, financing, and speculation. [1] Analogously, Pendle is a decentralized finance (DeFi) protocol designed to facilitate yield tokenization. It enables users to split their yield-bearing tokens into two separate tokens: Principal Tokens (PTs) and Yield Tokens (YTs). PTs represent the ownership of the underlying asset, while YTs represent the rights to the yield generated by the asset. The PT can be redeemed for the underlying token at maturity, while the YT entitles holders to the yield that would be generated over the contract's lifetime. In traditional finance terms, PTs resemble zero-coupon bonds, and YTs are akin to floating rate notes with daily payouts.

At any point during the contract, users can mint new PTs and YTs from the underlying asset or burn PTs and YTs to retrieve the underlying asset through the Pendle Automated Market Maker (AMM), with a small swap fee applied to these transactions. This fee does not apply when redeeming PTs for the underlying asset after the contract has matured. Additionally, holding YTs entitles the holder to the rewards generated by the underlying, providing an additional incentive for investors. This feature allows users to leverage yields and rewards through the liquidity and flexibility offered by Pendle without having to own the underlying asset. Pendle charges a fee on the yield generated by YTs and matured, unredeemed PTs.¹ [2]

A more detailed specification of standardized yield stripping, including the standardised yield (SY) token which is omitted here, can be found on the Pendle whitepapers, [3] and [4].

1.2 Objectives and Scope of the Study

We examine Liquid Restaking Tokens (LRTs) on the Pendle protocol. We intend to provide comprehensive insights into the dynamics of LRTs on Pendle, exploring their performance, correlations, and potential for arbitrage. Specifically we wish to answer the following questions:

- What drives fixed yields? Specifically, is there a notable correlation between the yields and the price of the token?
- Does liquidity influence performance, i.e., do less liquid pools tend to offer higher yields?
- To what extent are LRTs correlated with each other?
- How are yields correlated across different tenors of the same token?
- Can viable arbitrage strategies be identified based on observed yield patterns and token correlations?

To address these questions, we analyze a variety of contracts with different underlying tokens, tenors, and liquidity levels. To address these questions, we analyze a variety of contracts with different underlying tokens, tenors, protocol design, and liquidity levels. Specifically, we focus on Etherfi eETH (Jun/Sep/Dec), Zircuit eETH (Jun), Renzo ezETH (Sep/Dec), Puffer pufETH (Jun/Sep), Bedrock uniETH (Jun/Sep), and Kelp rsETH (Jun/Sep). It is important to note that June contracts are the only ones for which we have complete data, as they have already matured, whereas other contracts are still ongoing at the time of analysis.

¹At the time of writing, this fee is 3%.

Data for the YTs and PTs is sourced from the Pendle API. Underlying token price data is obtained from Yahoo Finance, and Total Value Locked (TVL) data is retrieved from DefiLlama. We use daily open data for prices.

2 Drivers of Fixed Yields

We first introduce some useful formulae and concepts. Due to the mechanism of minting and burning PT and YT for the underlying, we always have the PT/YT invariant:

$$PT(t) + YT(t) = \text{Underlying}(t)$$

which the Pendle AMM maintains via arbitrage. If we consider the same formula with prices in terms of the underlying, then we instead get:

$$PT_{ETH}(t) + YT_{ETH}(t) = 1 \quad (1)$$

which is more convenient to use as it removes the USD exposure of the underlying. Furthermore, we know that as we approach the expiration date, T , that the PT must converge in value to the underlying, since it can be redeemed, and the YT must converge to 0 since it has emitted all its yield. We therefore assume that

$$\lim_{t \rightarrow T} PT_{ETH}(t) = 1 \quad \text{and} \quad \lim_{t \rightarrow T} YT_{ETH}(t) = 0. \quad (2)$$

Now, since the price of the PT must be given by discounting the price of the underlying from the time to maturity, which is 1 in nominal ETH, we know the present value of the PT is:

$$PT_{ETH}(t) = e^{-r((T-t)/365)} \quad (3)$$

where r is the annualised fixed yield when buying at t . Consequently, we know the fixed yield is given by:

$$r = -\frac{365}{T-t} \ln(PT_{ETH}(t)). \quad (4)$$

Here, r is a function of time since the fixed yield is dependent on the price of the PT when buying it. Since we have the PT/YT invariant in (1), the variability in the PT's price is related to the fluctuations in the YT's price. That is to say, an increase in the value of the YT will decrease the price of the PT.

The canonical method of gauging the price of the YT is implied APY, defined as: [5]

$$\text{Implied APY} = \left[\left(1 + \frac{YT}{PT} \right)^{\frac{365}{T-t}} \right] - 1. \quad (5)$$

Implied APY is a forward-looking measure that reflects market expectations of the underlying asset's yield and any rewards (e.g., EigenLayer points) over the term of the contract. It captures consumer sentiment and predictions about future interest rates, risk premiums, and potential policy changes. The order book and bid-ask spread on Pendle are given in terms of

implied APY, which facilitates a common reference for trading YTs. It enables investors to compare the expected returns of different tokens and make informed decisions based on their risk tolerance and yield expectations.

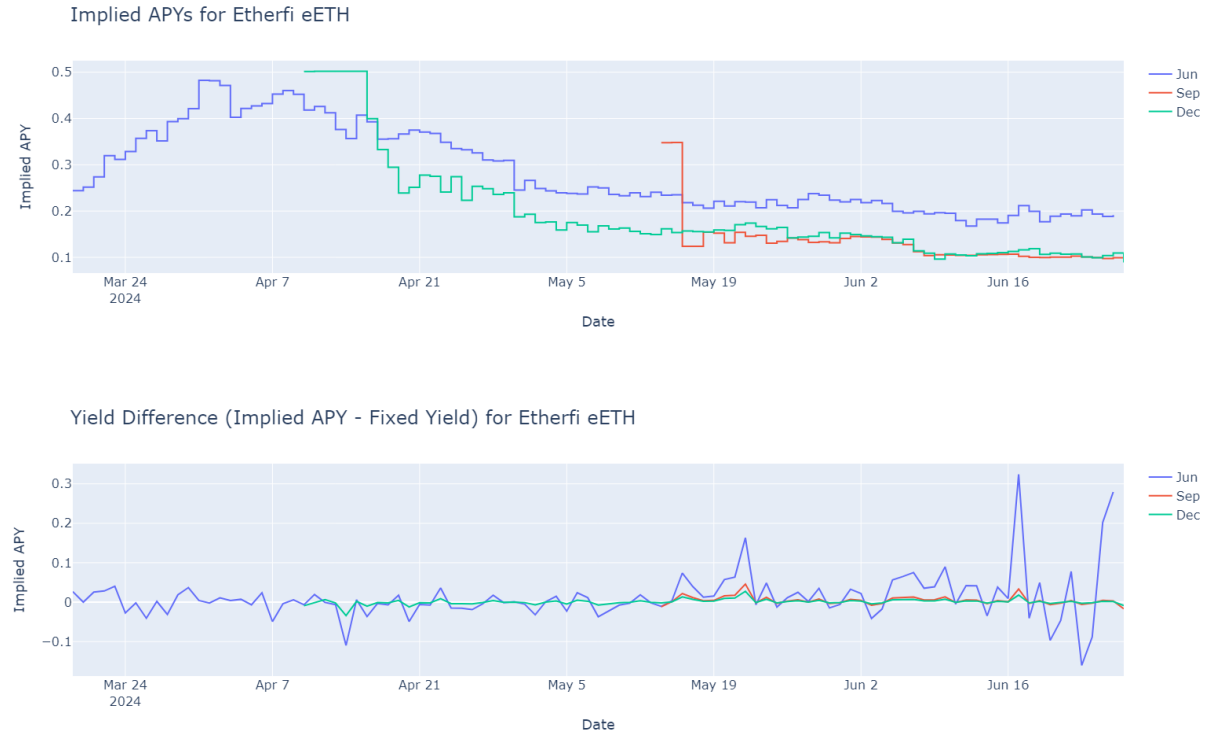


Figure 1: (Top) An example of the implied APY over time for Etherfi eETH contracts. By using the formula given in (5), we can replicate the graphs shown on Pendle. (Bottom) The difference between the implied APY and the fixed yield for Etherfi eETH contracts. The fixed yield formula used here assumes daily compounding to match with the formula used by Pendle for implied APY.

2.1 Correlation Analysis

We utilize correlation matrices as a qualitative tool to initially identify expected relationships and subsequently examine how our empirical data conforms to or deviates from these expectations. This approach aids in highlighting any anomalous or outlier behaviors that merit further investigation. Figure 2 showcases the correlation matrices for select contracts with different maturities and underlying tokens. Figures of the complete range of both the regular correlation matrices and correlation matrices in terms of nominal ETH can be found in Appendix A.

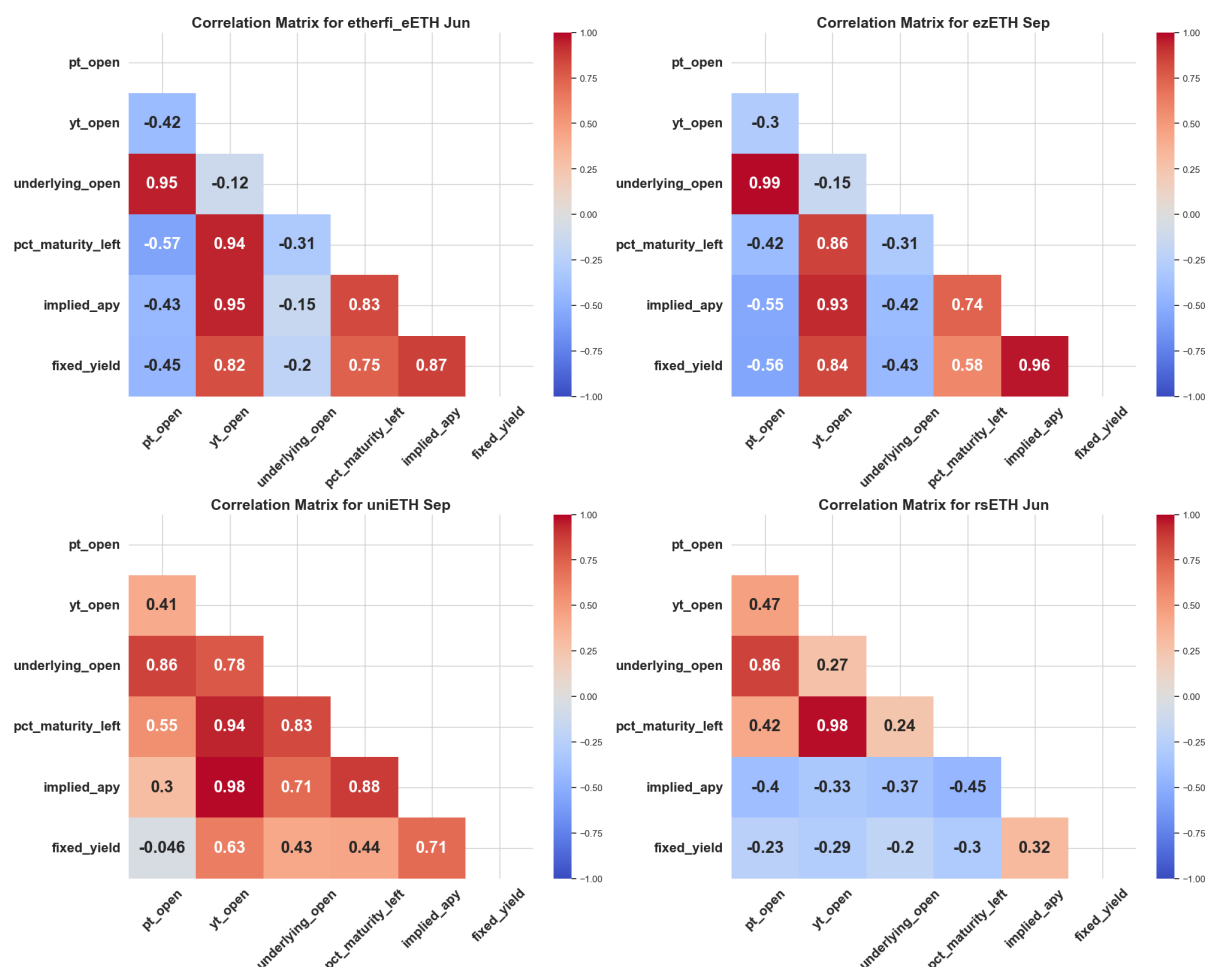


Figure 2: Correlation matrices of Etherfi eETH (Jun), ezETH (Sep), uniETH (Sep), rsETH (Jun). The diagonal and upper triangle have been hidden to preserve visual clarity.

2.1.1 Expected versus Observed Correlations

Each correlation discussed is first introduced with its theoretical expectation followed by an analysis based on the empirical data. The selected correlations focus on key aspects of PT and YT dynamics, underlying asset influences, and yield characteristics. Correlations not chosen had similar explanations or factors as those that were.

- **YT Open vs. PT Open:**

Expected Correlation: Strongly negative due to the YT/PT invariant. An increase to the price of YT should correspond to a perfect decrease in the price of PT should the underlying price remain constant.

Empirical Observation: Tokens with negative correlations were generally not as strong as expected. However, when looking at short-term (2 or 3 day) rolling windows of prices, the correlation becomes stronger. Similarly, looking at the correlation between the prices in nominal ETH makes an almost perfect negative correlation. Therefore, it is likely that noise is reducing the strength of the signal.

- **Underlying Open vs. PT Open:**

Expected Correlation: Positive correlation, since the price of PT is derived from the discounted price of the underlying.

Empirical Observation: The correlation is very strong across every token. For tokens with a slight weaker relationship (e.g., 0.85) there may be some uncertainty or market inefficiencies that warrant further investigation.

- **Maturity Left vs. PT Open and YT Open:**

Expected Correlation: and negative correlation with PT.

Empirical Observation: The correlation between duration and PT price is negative but not strongly established. When put into nominal ETH, the relationship becomes much stronger, suggesting the noise from the underlying is significant. However, there remain outliers. For example, for rsETH Jun, the correlation remains positive in both instances, suggesting the price of the PT decreases as we reach maturity.

- **Maturity Left vs. YT Open:**

Expected Correlation: Positive correlation with YT. This is because we know the price of the YT decreases over time as it emits yield.

Empirical Observation: Correlations between the duration of the contract and YT price are high across all tokens. The relationship is further reinforced by the data from correlation in nominal ETH.

- **Implied APY vs. YT Open:**

Expected Correlation: Strongly positive. The implied APY is meant as a way to gauge sentiment on future yields and consequently YT prices.

Empirical Observation: The correlation is strong across the board, suggesting implied APY is a good metric for investors to use. However, rsETH Jun is again an outlier with a negative correlation. Arguably, this is because implied APY is related to PT price which we have seen is abnormal for the contract; likely the two are related.

- **Maturity Left vs. Underlying Open:**

Expected Correlation: None expected. This should be considered noise as it depends on the performance of the underlying. Since all token prices are based on the price of ETH, there is likely some general market trends that are reflected.

Empirical Observation: Many tokens have negative correlations. Some have positive correlations. Considering this measures the evolution of the underlying token's price over the contract, it may give some idea of the noise affecting other correlations.

- **Implied APY vs. Maturity Left:**

Expected Correlation: Strongly positive. This is because the ratio between YT and PT decreases over time, which is used in the implied APY formula.

Empirical Observation: There is a strong positive correlation across all tokens apart from rsETH Jun which is once again an outlier.

- **Fixed Yield vs. Implied APY:**

Expected Correlation: Strongly positive. Suppose we work in nominal ETH. Then fixed yield increases when the PT price decreases. Similarly, implied APY increases when either PT decreases or YT increases. We have seen that PT and YT prices should have a negative correlation.

Empirical Observation: Many tokens have extremely high positive correlation between the two, as expected. Some tokens, like rsETH Jun, have a weak positive correlation while others, such as uniETH Jun, have a negative correlation. This hints at the possibility of arbitrage dynamics between PT and YT if they are not properly expressed.

2.1.2 Analysis of Anomalies

There are some tokens which deviate from theoretical expectations. Notably, rsETH Jun and uniETH Jun are good examples. Below is a comparison of PT and underlying prices of Etherfi eETH Jun and ezETH Jun. Etherfi eETH conforms well the theoretical ideas. The gap between the underlying and PT prices converges over time and almost vanishes at maturity. When standardised in terms of the underlying, the price follows a slow arc ending at 1.

On the other hand, the gap between the underlying and PT prices of rsETH Jun converges slower, with a significant gap still remaining near maturity. This leads to large fixed yield values at the time since one can buy a PT at approximately 0.97 of the underlying and redeem it the next day to generate large time-adjusted yields. Furthermore, there is a period around May 26th where the underlying crashes and the price of the PT is temporarily higher than the price of the underlying. Consequently, the price of the yield token would also have to be negative to maintain the PT/YT invariant which is nonsensical. Likely these anomalies are causing unexpected correlations.

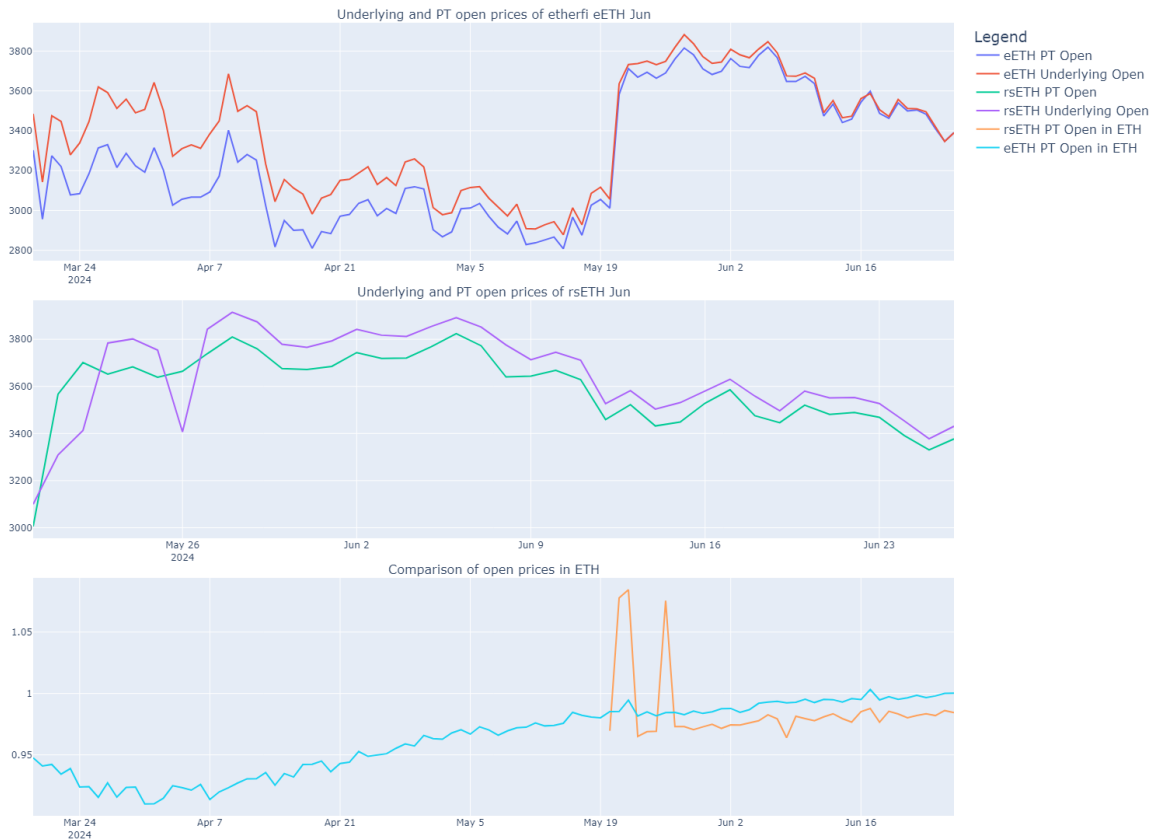


Figure 3: (Top) Underlying and PT open prices of Etherfi eETH Jun. (Middle) Underlying and PT open prices of rsETH Jun. (Bottom) A comparison of the PT open prices in nominal ETH between the two.

Below are the correlation matrices of Etherfi eETH Jun, uniETH Jun, and rsETH Jun respectively. Etherfi eETH Jun is the prototypical matrix one would expect in line with the dynamics of pricing PT and YT. If the matrix deviates from this realm of correlations, then there is very likely to be market inefficiencies that can be exploited by arbitrage strategies. For example, if the price of PT is higher than the underlying, as seen in the previous example, then one could buy and underlying, swap it into a PT and YT, and sell the PT while keeping the YT. This would net a profit as long as the gain is greater than the swap fees. We will discuss arbitrage ideas in detail later.

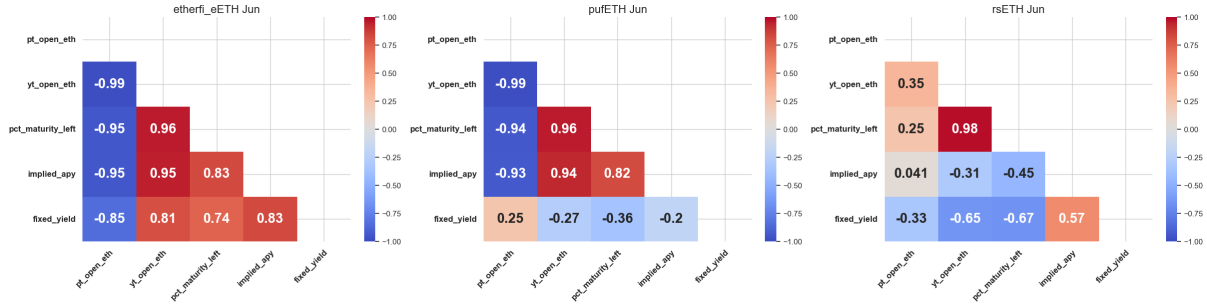


Figure 4: Correlation matrices of (Left) Etherfi eETH Jun, (Middle) uniETH Jun, (Right) rsETH Jun in nominal ETH.

2.2 Principle Component Analysis

In order to understand the drivers of fixed yield and PT prices comprehensively, we employ Principal Component Analysis (PCA). By reducing the dimensionality of our dataset, PCA helps to uncover the principal components that account for the most variance in the data. Through PCA, we aim to isolate the key features that influence PT prices and fixed yields, providing clearer insights into how these elements interact and their overall impact on market behavior.

2.2.1 Fundamental Characteristics

The first PCA looks into the fundamental characteristics discussed previously. Namely, opening prices, implied APY, fixed yield, and the contract's remaining duration. We further solidify the interaction between these.

Table 1: Loadings of the Principal Components for Fundamental Components

	PT Open	YT Open	Underlying Open	Maturity Left	Implied APY	Fixed Yield
PC1	-0.5378	0.4839	-0.4237	0.4015	0.3411	-0.1401
PC2	-0.3237	-0.3298	-0.4736	-0.5326	0.1972	0.4895

The PCA reveals that the first principal component explains approximately 45.49% of the variance, while the second accounts for 22.34%. The loadings of the first principal component suggest that the YT open, percentage maturity left, and implied APY have positive coefficients, indicating a common trend where longer maturities and higher anticipated yields drive the YT prices up. This aligns with theoretical expectations where longer duration encompass more yield potential, reflected in higher YT prices. The negative coefficients for PT open and underlying open in the first principal component indicate that increases in YT values, due to higher yields and longer maturities, inversely affect PT prices. This conforms with the concept of the

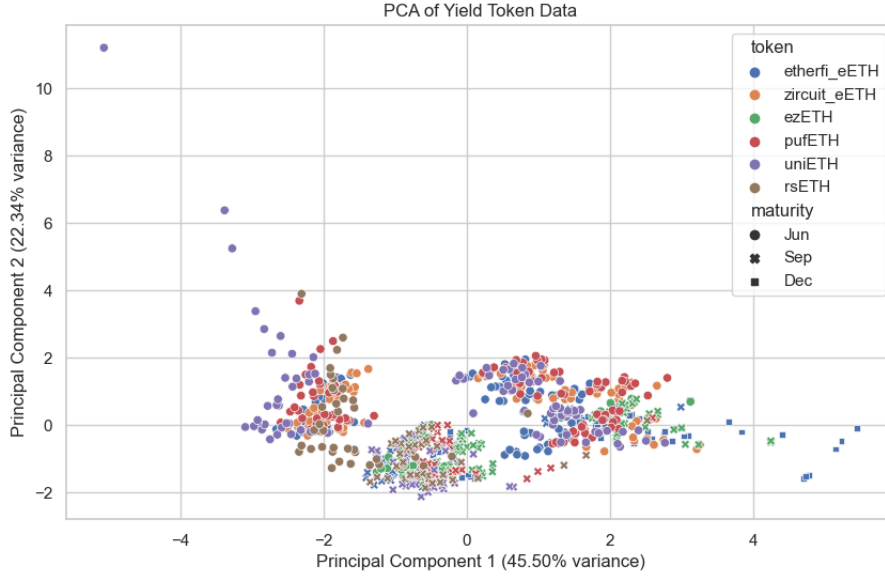


Figure 5: A PCA using PT open, YT open, underlying open, maturity of the contract, implied APY, and fixed yield.

PT/YT invariant. Overall, the loadings of this component support the theoretical pricing model discussed.

The second principal component explains an additional 22.34% of the variance in the dataset. Maturity left shows a strong negative relationship with PC2, highlighting this component might best represent market conditions as the contract reaches expiry. PT Open, YT Open, and Underlying Open exhibit negative coefficients in this component, suggesting that these prices tend to decrease when PC2 values increase. This might be indicative of market dynamics as contracts approach maturity, where the value of the principal tokens and their underlying assets might drop due to increased market volatility or risk assessments. This pattern could reflect the escalating uncertainty or speculative trading as fewer days remain, potentially leading to more drastic price movements in PT and YT. Conversely, Implied APY has a positive coefficient, albeit smaller, suggesting that higher expected yields could correlate with certain market conditions that influence PC2 to rise, such as the time near expiry or low underlying prices. This aspect underscores the sensitivity of market dynamics to yield expectations, particularly as contracts mature. The most significant positive loading is observed in Fixed Yield, indicating that higher fixed yields are associated with an increase in PC2. This relationship points to the market's response to better yield prospects, which might be attracting more trading activity or influencing pricing strategies as maturity nears.

Recall the yield difference graph shown in Figure 1. Intuitively, it appears that the volatility in the difference is higher towards the end of the contract, which would be in accordance with the relationships in PC2. We confirm this by splitting the data into an early period and late period (with 10% of the contract to go) and using Leven's test on the June contracts. The following results are obtained:

Table 2: Levene’s Test Results for Volatility in the Last 10% of the Contract Duration

Token	Statistic	p-value
Etherfi eETH	75.34	8.80×10^{-14}
Zircuit eETH	70.22	6.24×10^{-13}
pufETH	11.97	0.0008
uniETH	9.84	0.0027
rsETH	11.61	0.0016

These statistically significant results suggest that the volatility of the yield difference increases as the contract approaches maturity. This heightened volatility near the end of the contract could potentially create arbitrage opportunities. However, it is important to note that these findings might also be influenced by noise and prevailing market conditions. Therefore, additional data is required to confirm these observations and assess the consistency of this pattern across different market environments.

2.2.2 Volatility

2.2.3 Trading Volumes

Liquidity data for PT/YT token pairs on Pendle is not publicly available. We use the logarithm of the trading volume of the PT, YT, and underlying token as a proxy for liquidity. Since the trading volume on some days is 0, presumably due to data not being available, we take a 5-day moving average.

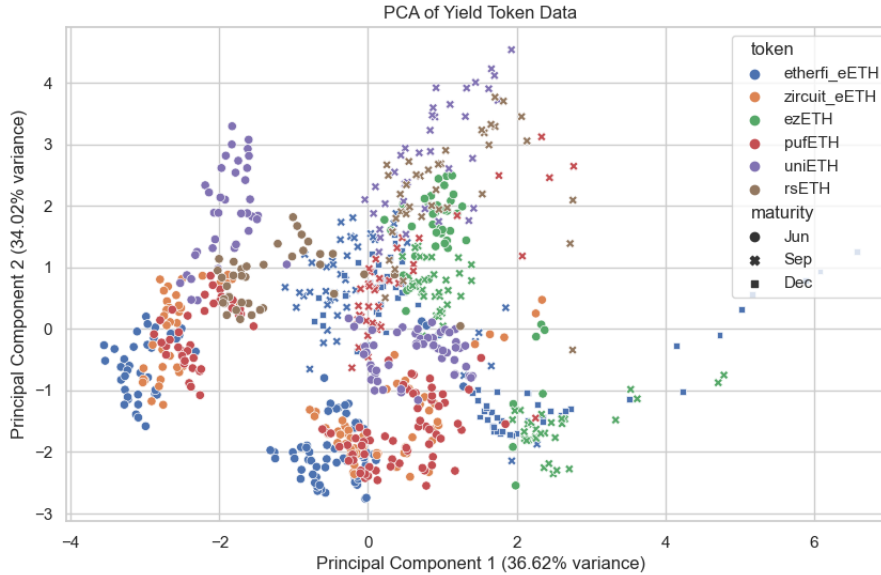


Figure 6: A PCA using PT open, YT open, underlying open, maturity of the contract, and 5-day moving averages of the log volume.

The first principal component, accounting for approximately 36.62% of the total variance, highlights major trends in trading behavior and price movements. Lower PT open prices negatively correlate with this component, suggesting increased market activity when prices dip, possibly as investors capitalize on lower prices. Conversely, higher YT open prices positively impact this component, indicating increased investor interest or speculative activity as these prices

Table 3: Loadings of the Principal Components for Volumes

	PT Open	YT Open	Underlying Open	PT Volume	YT Volume	Underlying Volume	Maturity Left
PC1	-0.3927	0.5109	-0.2216	-0.3260	-0.3003	0.2064	0.5446
PC2	0.4538	-0.0833	0.5370	-0.4951	-0.4995	-0.0211	0.0601

rise. A similar negative association is observed with the underlying open prices, where lower prices might signal broader market corrections or bearish sentiments. The trading volumes for PT and YT exhibit negative loadings, implying that higher volumes often coincide with lower token prices, indicative of selling pressure or phases of liquidation. The percentage of maturity left shows a strong positive loading, suggesting longer contract duration correlates with higher component values, reflecting higher risk or yield expectations from longer contracts.

The second principal component, explaining an additional 34.02% of the variance, offers insights into more nuanced behaviors. It highlights that PT and underlying open prices have large positive loadings, suggesting bullish market conditions or strong buying interest when these prices increase. In contrast, YT open prices have a minimal negative loading, indicating a slight inverse relationship and suggesting that increases in PT and underlying prices may coincide with less YT trading. This component focuses on a case where contract maturity and underlying volume impacts are relatively negligible, focusing more on immediate trading dynamics rather than long-term time evolution. PT and YT volumes have significant negative loadings, reinforcing the idea that periods of large trading volumes may depress prices due to market saturation or high liquidity or be due to crashes in the underlying token's price.

The issue with using volumes as a proxy for liquidity is that it is not immediately obvious whether high volume indicates large sell-offs and price fluctuations or stability in price due to arbitrage bots and liquidity providers. We demonstrate this by looking at the correlation between trading volume and price volatility.

2.2.4 TVL

While we will look at the effect of the underlying token's TVL on yields later, we here introduce it to show its role as a stabilizing factor in price relationships and its utility in clustering tokens.

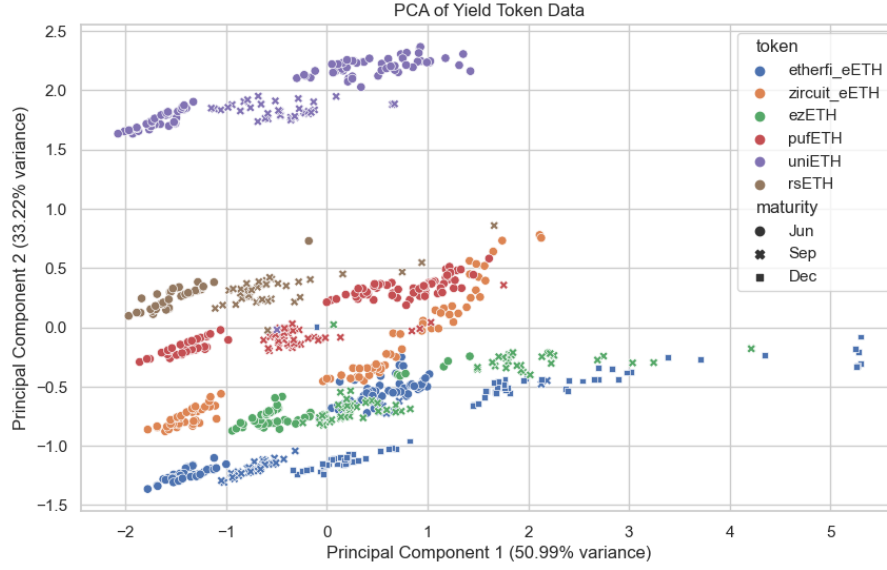


Figure 7: A PCA using PT open, YT open, and log TVL. The addition of log TVL creates strata of tokens which may be useful in clustering algorithms. The addition of other features can enhance isolation.

The first principle component, explaining 50.99% of the variance, shows a strong inverse relationship between PT and YT open. This indicates that when PT prices increase, YT prices tend to decrease and vice versa, supporting the PT/YT invariant concept where shifts in one are counterbalanced by the other. The smaller positive loading of log TVL suggests that tokens with higher TVL show a less pronounced inverse relationship between PT and YT prices, possibly indicating more market stability or confidence in tokens with higher liquidity.

The second principle component, explaining 33.22% of the variance, is dominated by the negative loading of log TVL, indicating that variations in TVL are a significant factor distinguishing tokens at this component level. The negative loading implies that tokens with lower TVL might be subject to more volatility or less price stability in PT prices. The minimal loadings on PT on YT open suggest that, at this level, the stability impact of TVL is relatively small.

2.3 Random Forest Regression and Fourier Analysis

3 Liquidity Impact of the Underlying

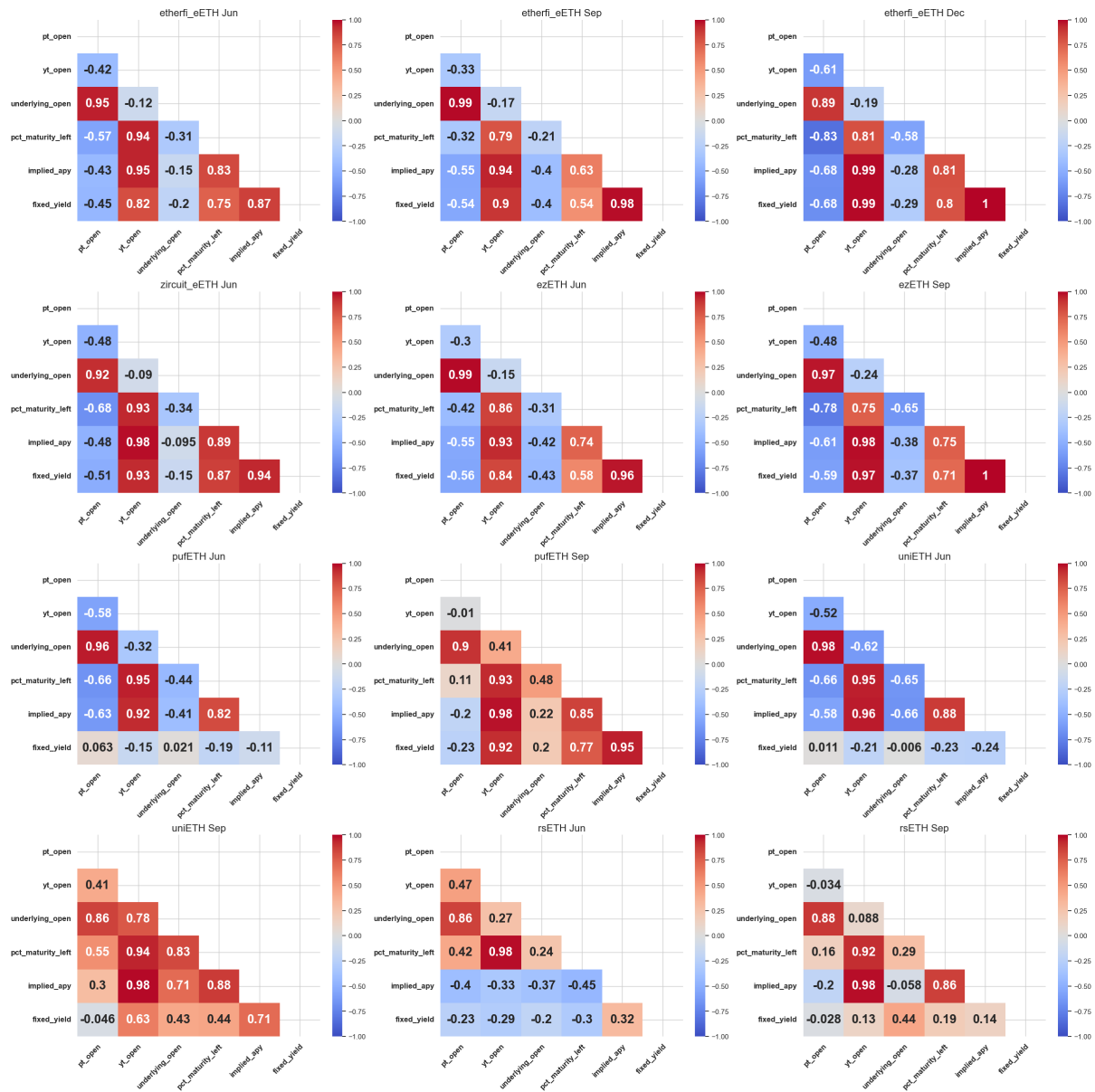
In many financial settings, illiquid instruments offer higher yields for the risks they come with. For example, corporate bonds have higher yields than treasury bonds due to a company's risk of defaulting. We attempt to quantify the existence of a similar effect by considering whether the TVL for the underlying token impacts the respective YT and PT. In particular, we use the quantiles of log TVL to partition groups of tokens and verify if there is an implicit difference in the fixed yield and implied APY. Here, quantile 0.0 represents the lowest TVL (least liquid pools) and quantile 3.0 represents the highest TVL (most liquid pools).

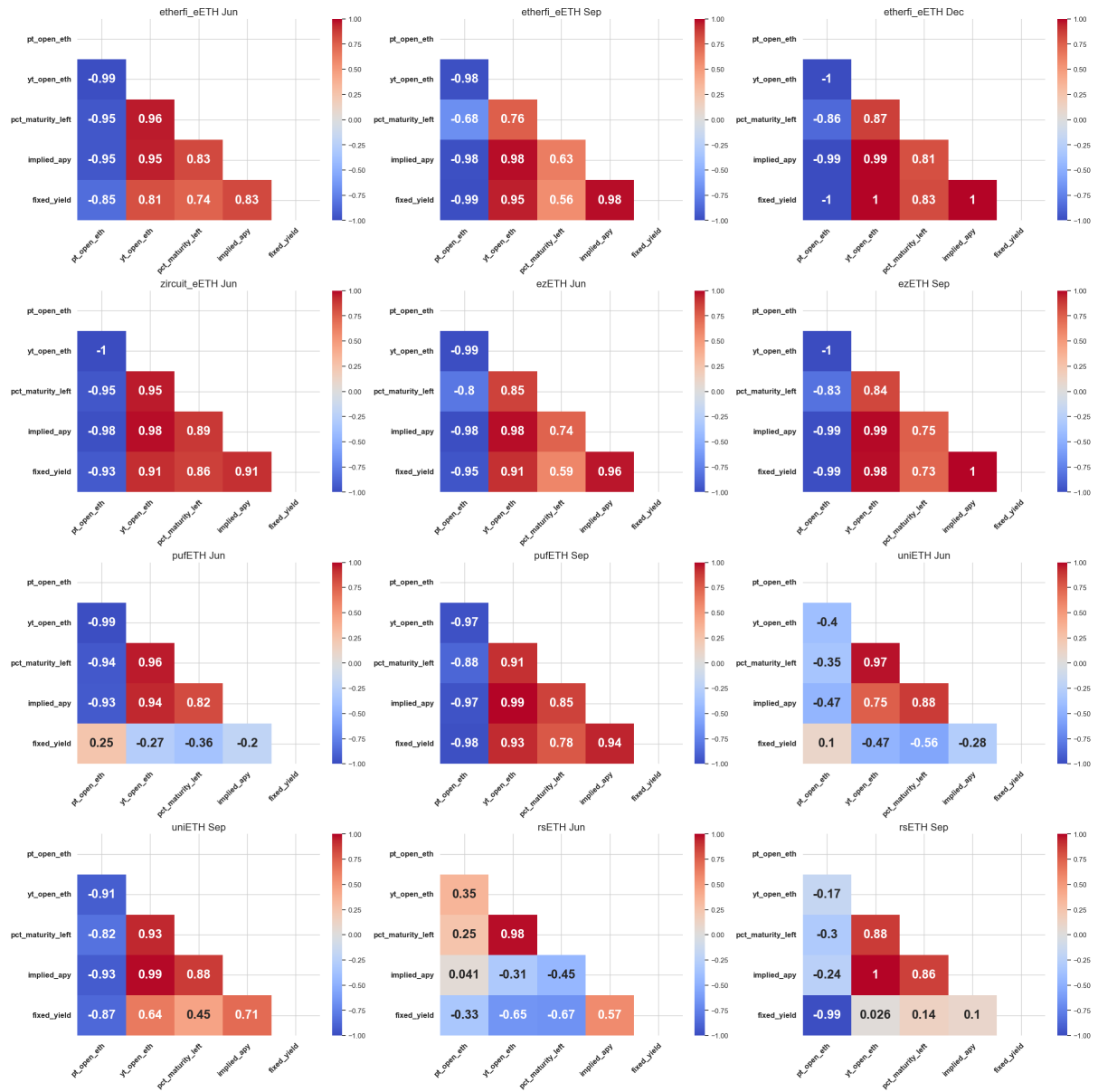
4 Concluding Remarks

4.1 Summary of Results

4.2 Further Areas of Study

A Complete Correlation Matrices





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